Lexmi Nicos Order Suite

"An elegant interface for seamless order placement"

Submitted by:
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in partial fulfillment for the award of the

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SUGUNA PIP SCHOOL COIMBATORE

BONAFIDE CERTIFICATE

Certified that this project report **Lexmi Nicos Order Suite** is the bonafide work of **DEVEHH SURESH** who carried out the project work under my supervision.

SIGNATURE OF EXTERNAL EXAMINER SIGNATURE OF INTERNAL EXAMINER

SIGNATURE OF PRINCIPAL

PROJECT REPORT - LEXMI NICOS ORDER SUITE

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DECLARATION

I, DEVESHH SURESH, of Class 12 A1, hereby declare that the project titled Lexmi Nicos Order Suite, submitted to Suguna PIP School (Affiliation Number 1930213), Nehru Nagar, Kalapatti Road, Coimbatore - 641014, with regard to the Internal Examination for Class 12, is a record of original project work carried out by me under the supervision of Mrs. Nithya R, faculty member of the Computer Science Department, Suguna PIP School, Coimbatore.

I hereby certify that the work contained in this report is original and has been completed by me under her supervision

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ABSTRACT

This project presents the **Lexmi Nicos Order Suite**, a sophisticated, **multi-tiered digital ordering system** developed using **Python's Tkinter** library for the Graphical User Interface (GUI) and **MySQL** for secure data persistence.

The application's central elegance lies in its **stateful and modular architecture**, which seamlessly guides users through a complex transaction process. The system begins with a critical **authentication layer**, differentiating access between staff and guests, and providing specific verification for members against a connected MySQL database.

The core functionality involves a dynamically linked ordering process across Appetizers, Entrees, and Desserts. Crucially, the system employs a synchronized network of global variables that ensures the master order list is **updated in real-time** across all navigation screens upon submission. This persistent data model culminates in a **Final Order Summary**, which not only presents a definitive review of the customer's selections but also incorporates an **intuitive error-correction loop**, allowing instant navigation back to any category for modification. The Lexmi Nicos Order Suite stands as a demonstration of effective, user-centric application design, unifying a sleek front-end experience with a robust, database-driven back-end logic.

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INTRODUCTION

Welcome to the **Lexmi Nicos Order Suite**, a system architected for operational precision, dedicated to elevating the efficiency and reliability of the front-of-house ordering experience. This suite provides a streamlined, rigorously authenticated pathway from client identification to final order compilation. The project's central aim is to deliver a robust, error-proof solution that eliminates transactional ambiguity and synchronizes real-time data flow across all user interfaces.

This sophisticated application is constructed upon a core foundation of Python and key external modules, each fulfilling a vital function in the system's integrity:

Module	**Primary Function**
tkinter	**Graphical User Interface (GUI)**: Utilized to construct the multi-window environment, including the dynamic root, authentication screens (wind5, wind6), main menu (wind3), and category interfaces.
mysql.connector	**Database Integration**: Essential for establishing and managing the connection to the lexminicos database,

	enabling secure **Member and Staff Authentication** via credential lookup.
tk.StringVar	**State Management & Data Capture**: Employed locally within category windows (e.g., Appetizers) to temporarily bind and capture precise quantity inputs from the entry fields before the final submission and global update.

SYSTEM FEATURES AND OPERATIONAL FUNCTIONALITIES

The **Lexmi Nicos Order Suite** is defined by its structured workflow and critical functional components:

- **Multi-Tiered Authentication:** The system provides distinct access control for **Staff** (via staff() function) and **Guests** (via guest() function), further segmenting guests into **Members** (requiring passcode verification) and non-members for tailored service.
- **Persistent Order State:** Order data is managed using **global list variables** (appsorder, entreesorder, dessertsorder). This ensures that selections are continuously tracked and never lost when navigating between the Appetizer, Entree, and Dessert windows.
- **Real-Time Synchronization:** The **Submit** command in each category window triggers a critical synchronization function, which updates the global lists and instantaneously rebuilds the master order list, displaying the complete order live within the current category screen.
- **Intuitive Error Correction Loop:** The **finaliseorder()** window presents the complete, final selection summary while retaining all category navigation buttons. This design allows users to spot an error, jump back to the specific

category, correct the quantity, resubmit, and return to the final summary with changes immediately reflected.

- **Dynamic Menu Presentation:** Item lists are dynamically bound to entry fields, allowing users to input exact quantities for any of the 30 available items across the three primary categories.

SOURCE CODE

```
import tkinter
import mysql.connector
tk = tkinter

mycon = mysql.connector.connect(host="localhost", user="root", password="Suresh23",
database="lexminicos")
cursor = mycon.cursor()

import mysql.connector
#----------MYSQL------------------
mycon = mysql.connector.connect(host="localhost",
user="root",password="Suresh23",database="lexminicos" )
cursor = mycon.cursor()

def fetch_members():
    """Fetches member name and password from the 'members' table and returns them as a dictionary."""
    cursor.execute("SELECT name, password FROM members")
    member_data = dict(cursor.fetchall())
    return member_data

def fetch_staff():
    """Fetches staff name and password from the 'staff' table and returns them as a dictionary."""
    cursor.execute("SELECT name, password FROM staff")
    staff_data = dict(cursor.fetchall())
    return staff_data
# ---------- MAIN WINDOW ----------
```

```
#BLOCK 1 CRETE THE HOMEEPAGE OR STARTING PAGE,    BLOCK 2 SHOWS MENU
BLOCK 3 ASKS IF MEMBER OR NOT    BLOCK 4 IS IF STAFF
root = tk.Tk()
root.title('MENU')
root.geometry('1500x600')
root.configure(bg='black')

# global variables for selections
s = ''  # will store final order as string
i = tk.IntVar()  # variable for member/guest selection
i.set(3)  # initial value

final = []  # final combined order list
appsorder = []  # order quantities for appetizers
entreesorder = []  # order quantities for entrees
dessertsorder = []  # order quantities for desserts

# ---------- BLOCK 2 FUNCTION TO SHOW MENU BASED ON MEMBER/GUEST ----------
def showmenu():
    # if a choice has been made (member or not)
    if i.get() != 3:
        global wind2
        wind2.withdraw()  # hide the guest/member selection window

    # if not a member
    if i.get() == 0:
        discount = 0  # placeholder for discounts if needed
        menu()  # open main menu
    else:
        # member login window
        global wind5
        wind5 = tk.Toplevel(root)
        wind5.title('Member Verification')
        wind5.geometry('600x300')
        wind5.configure(bg='black')
```

```
    # Name Label and Entry
    namelabel = tk.Label(wind5, text='Name:', fg='white', bg='black', font=('Arial', 12))
    namelabel.place(x=50, y=40)
    name = tk.Entry(wind5, width=30, bg='white', font=('Arial', 12))
    name.place(x=150, y=40)

    # Passcode Label and Entry
    passcodelabel = tk.Label(wind5, text='Passcode:', fg='white', bg='black', font=('Arial', 12))
    passcodelabel.place(x=50, y=90)
    passcode = tk.Entry(wind5, width=30, bg='white', show='*', font=('Arial', 12))
    passcode.place(x=150, y=90)

    members = fetch_members() # Fetches data from MySQL

    # check login credentials
    def check():
      namegiven = name.get()
      code = passcode.get()
      a=0
      for i in members.keys():
        if namegiven in members.keys():
          if code == members[namegiven] and i==namegiven:
            a=a+1
      if a==0:
        result = tk.Label(wind5, text='ACCESS DENIED', fg='white', bg='black', font=('Arial', 12,
'bold'))
        result.place(x=220, y=180)
      else:
        menu()

    # Submit button to verify member login
    submit = tk.Button(wind5, text='SUBMIT', font=('Arial', 12, 'bold'),
                fg='black', bg='#E16C2E', width=10, height=1, command=check)
    submit.place(x=250, y=140)
```

```
# ---------- BLOCK 3 FUNCTION FOR GUEST SELECTION ----------
def guest():
    global i, wind2
    root.withdraw()  # hide main root window
    wind2 = tk.Toplevel(root)
    wind2.geometry('1500x600')
    wind2.configure(bg='black')

    # Radio button for member
    member = tk.Radiobutton(wind2, text='member', font=(30), fg='black', bg='#E16C2E', value=1,
variable=i)
    member.place(x=50, y=100)

    # Radio button for non-member
    notmember = tk.Radiobutton(wind2, text='not a member', font=(30), fg='black', bg='#E16C2E',
value=0, variable=i)
    notmember.place(x=300, y=100)

    # Submit button to continue
    submit = tk.Button(wind2, text='SUBMIT', font=(30), fg='black', bg='#E16C2E',
command=showmenu)
    submit.place(x=175, y=300)

    # Instructions button
    how = tkinter.Button(wind2, text='INSTRUCTIONS', font=('Arial', 14), fg='black',
bg='#E16C2E', command=lambda: [wind2.withdraw(), howto()])
    how.place(x=500, y=400)

# ---------- BLOCK 4 FUNCTION FOR STAFF SELECTION ----------
def staff():
    root.withdraw()
    wind6 = tk.Tk()
    wind6.geometry('600x300')
    wind6.title('Staff Verification')
    wind6.configure(bg='black')
```

```
  # Name label and entry
  namelabel = tk.Label(wind6, text='Name:', fg='white', bg='black', font=('Arial', 12))
  namelabel.place(x=50, y=40)
  name = tk.Entry(wind6, width=30, bg='white', font=('Arial', 12))
  name.place(x=150, y=40)

  # Passcode label and entry
  passcodelabel = tk.Label(wind6, text='Passcode:', fg='white', bg='black', font=('Arial', 12))
  passcodelabel.place(x=50, y=90)
  passcode = tk.Entry(wind6, width=30, bg='white', show='*', font=('Arial', 12))
  passcode.place(x=150, y=90)

  employee = fetch_staff() # uses my sql


  # check login credentials
  def check():
    namegiven = str(name.get())
    code =str(passcode.get())
    a=0
    for i in employee.keys():
      if i==namegiven and code == employee[i]:
#if namegiven not even in employee defenitely the 2nd condition will not be checked so then no error
        a+=1
    if a==0:
      result = tk.Label(wind6, text='ACCESS DENIED', fg='white', bg='black', font=('Arial', 12,
'bold'))
      result.place(x=220, y=180)
    else:
      result = tk.Label(wind6, text='ACCESS GRANTED', fg='white', bg='black', font=('Arial', 12,
'bold'))
      result.place(x=220, y=180)

  # Submit button to verify staff login
  submit = tk.Button(wind6, text='SUBMIT', font=('Arial', 12, 'bold'), fg='black', bg='#E16C2E',
width=10, height=1, command=check)
```

```
    submit.place(x=250, y=140)

# ---------- STAFF AND GUEST BUTTONS IN MAIN WINDOW FROM BLOCK 1 ----------
staff_btn = tk.Button(root, text='click if staff', font=(30), fg='black', bg='#E16C2E', command=staff)
staff_btn.place(x=50, y=100)

guest_btn = tk.Button(root, text='click if guest', font=(30), fg='black', bg='#E16C2E',
command=guest)
guest_btn.place(x=200, y=100)

# ---------- FOOD LISTS ----------
appslist = [
    "Spinach Artichoke Dip", "Crispy Spring Rolls", "Caprese Skewers", "Mini Crab Cakes",
    "Garlic Parmesan Breadsticks", "Loaded Potato Skins", "Shrimp Cocktail",
    "Hummus Platter", "Chicken Satay", "Bruschetta"
]

entreeslist = [
    "Grilled Salmon", "Chicken Tikka Masala", "Classic Beef Lasagna", "Mushroom Risotto",
    "Pan-Seared Duck Breast", "Spicy Shrimp Scampi", "Vegetable Korma",
    "New York Strip Steak", "Lamb Shank", "Black Bean Burger"
]

dessertslist = [
    "Chocolate Lava Cake", "New York Style Cheesecake", "Tiramisu", "Apple Crumble",
    "Crème Brûlée", "Brownie Sundae", "Key Lime Pie", "Red Velvet Cake",
    "Sticky Toffee Pudding", "Mango Mousse"
]

# ---------- INSTRUCTIONS WINDOW ----------
def howto():
    global howwind
    howwind1 = tk.Toplevel(root)
    howwind1.geometry('1500x900')
    howwind1.configure(bg='black')
    howwind1.title('How to Use the App')
```

```
    instructions = (
        "📝 HOW TO USE:\n\n"
        "• If you're a member, select 'Member' and enter any text for Name and Passcode, then click 'Login'.\n"
        "• If you're not a member, choose 'Not a Member' and click 'SUBMIT' to open the main 'LEXMI NICOS' menu.\n\n"
        "• On the menu screen, click 'Appetizers', 'Entrees', or 'Desserts' to browse food categories.\n"
        "• In each category, type the quantity you want next to each item (leave it blank or 0 if you don’t want that item).\n"
        "• Important: click 'Submit' at the top-left to save your selections before moving to another category.\n\n"
        "• Use the 'Go to' buttons to navigate between categories.\n"
        "• Click 'Back to Menu' anytime to return to the main menu.\n"
        "• Once done, click 'Finalise Order' to review your complete order.\n"
        "• To make changes, go back using the 'Go to' buttons, edit quantities, click 'Submit' again, and re-finalize your order.\n\n"
        "• To exit, simply close the main 'MENU' window using the 'X' button.\n\n"
        "💡 Tips:\n"
        "  - Always click 'Submit' after entering quantities.\n"
        "  - Use only whole numbers (no decimals or letters).\n"
        "  - This version does not calculate prices."
    )

    h = tk.Label(
        howwind1,
        text=instructions,
        font=('Arial', 14),
        fg='#E16C2E',
        bg='black',
        justify='left',
        wraplength=1450
    )
    h.pack(padx=30, pady=30, anchor='w')

    back = tk.Button(
```

```
        howwind1,
        text='BACK',
        font=('Arial', 14, 'bold'),
        width=25,
        bg='#E16C2E',
        fg='black',
        command=lambda: [howwind1.destroy(), wind2.deiconify()]
    )
    back.pack(pady=20)

# ---------- MENU WINDOW ----------
def menu():
    global wind3
    wind3 = tkinter.Toplevel(root)
    wind3.title('MENU')
    wind3.geometry('1500x600')
    wind3.configure(bg='black')

    wlcm = tkinter.Label(wind3, text="WELCOME TO", font=('TIMES NEW ROMAN', 12),
fg='white', bg='black')
    wlcm.place(x=740, y=0)

    ttl = tkinter.Label(wind3, text="LEXMI NICOS", font=('ELEPHANT', 40), fg='#E16C2E',
bg='black')
    ttl.place(x=570, y=20)

    # Category buttons
    appsbutton = tkinter.Button(wind3, text="Appetizers", font=('Arial', 14), fg='black',
bg='#E16C2E', width=15, height=2, command=apps)
    appsbutton.place(x=50, y=120)

    entreesbutton = tkinter.Button(wind3, text="Entrees", font=('Arial', 14), fg='black', bg='#E16C2E',
width=15, height=2, command=entrees)
    entreesbutton.place(x=250, y=120)
```

```
  dessertsbutton = tkinter.Button(wind3, text="Desserts", font=('Arial', 14), fg='black',
bg='#E16C2E', width=15, height=2, command=desserts)
  dessertsbutton.place(x=450, y=120)

# ---------- FINAL ORDER WINDOW ----------
def finaliseorder():
  global finalwind
  finalwind = tkinter.Toplevel(root)
  finalwind.title('FINAL ORDER')
  finalwind.geometry('1500x900')
  finalwind.configure(bg='black')

  title_label = tk.Label(finalwind, text="YOUR ORDER", font=('ELEPHANT', 20), fg='#E16C2E',
bg='black')
  title_label.place(x=600, y=20)

  # Buttons to go back to categories or main menu
  apps_button = tk.Button(finalwind, text="Go to Appetizers", font=('Arial', 12), fg='black',
bg='#E16C2E', command=lambda: [finalwind.withdraw(), apps()])
  apps_button.place(x=150, y=20 * (len(dessertslist) + 6))

  entrees_button = tk.Button(finalwind, text="Go to Entrees", font=('Arial', 12), fg='black',
bg='#E16C2E', command=lambda: [finalwind.withdraw(), entrees()])
  entrees_button.place(x=300, y=20 * (len(dessertslist) + 6))

  desserts_button = tk.Button(finalwind, text="Go to Desserts", font=('Arial', 12), fg='black',
bg='#E16C2E', command=lambda: [finalwind.withdraw(), desserts()])
  desserts_button.place(x=450, y=20 * (len(dessertslist) + 6))

  back_button = tk.Button(finalwind, text="Back to Menu", font=('Arial', 12), fg='black',
bg='#E16C2E', command=lambda: [finalwind.withdraw(), menu()])
  back_button.place(x=600, y=20 * (len(dessertslist) + 6))

  # prepare final order list
  global final
  final = []
```

```
    for i in range(len(appslist)):
        if i < len(appsorder) and appsorder[i] != "0":
            final.append(appslist[i] + ", " + appsorder[i])

    for i in range(len(entreeslist)):
        if i < len(entreesorder) and entreesorder[i] != "0":
            final.append(entreeslist[i] + ", " + entreesorder[i])

    for i in range(len(dessertslist)):
        if i < len(dessertsorder) and dessertsorder[i] != "0":
            final.append(dessertslist[i] + ", " + dessertsorder[i])

    # display final order
    for i in final:
        j = tk.Label(finalwind, text=i, font=(10), fg='white', bg='black')
        j.place(x=0, y=60 + 25 * (final.index(i)))

# ---------- APPETIZERS WINDOW ----------
def apps():
    global finalwind
    wind3.withdraw()
    appswind = tkinter.Toplevel(root)
    appswind.title('APPETIZERS')
    appswind.geometry('1500x900')
    appswind.configure(bg='black')
    global tapps
    tapps = []
#----------FINDING THE DISH WITH THE BIGGEST NAME-------------------------------
    e = ''
    for i in appslist:
        if len(i) > len(e):
            e = i
#------------------------------------------------------------------------------
    # create labels and entry fields for each appetizer
    for i in appslist:
        entry = tk.StringVar()
```

```
    entry.set('0')

    j = tk.Label(appswind, text=i, font=('Arial', 12), fg='white', bg='black')
    j.place(x=0, y=20 * appslist.index(i))

    k = tk.Entry(appswind, textvariable=entry, width=5, font=('Arial', 12))
    k.place(x=7*len(e)+100, y=20 * appslist.index(i))
    #THE ENTRY BOXES TO BE PLACED PERFECTLY USE THE LENGTH OF THE LONGEST DISHES NAME ALDREADY CALCULATED

    tapps.append(entry)

    '''NOTE: here it is like a list of lists, to explain it,
conside a list L=[a,b,c,d] here the letters correspond to the value stored in the entries and tapps is L.
now when the user changes the entry  in tapps,
the value a is changed, so that way when submit is hit we can et the values'''

  # function to save selections
  def appsfunc():
    global appsorder, final, entreesorder, dessertsorder
    appsorder = []
    for i in tapps:
      appsorder.append(i.get())

    final = []
    for i in range(len(appslist)):
      #since the tapps has entries stored as strings i am checking which and all in apps order have non 0 entries and put themalong with dish name into final

      if i < len(appsorder) and appsorder[i] != "0":
        final.append(appslist[i] + ", " + appsorder[i])

    for i in range(len(entreeslist)):
      if i < len(entreesorder) and entreesorder[i] != "0":
        final.append(entreeslist[i] + ", " + entreesorder[i])
```

```
    for i in range(len(dessertslist)):
        if i < len(dessertsorder) and dessertsorder[i] != "0":
            final.append(dessertslist[i] + ", " + dessertsorder[i])

    # display current selections
    global s
    s = ''
    for i in final:
        s = s + '\n' + i
    ordertext = tk.Label(appswind, text=s, bg='black', fg='white', font=(12))
    ordertext.place(x=800, y=0)

  Submit = tk.Button(appswind, text='Submit', font=('Arial', 14), fg='black', bg='#E16C2E',
command=appsfunc)
  Submit.place(x=0, y=20*(len(appslist) + 3))

  # buttons to navigate categories
  def open_entrees():
    appswind.withdraw()
    entrees()

  def open_desserts():
    appswind.withdraw()
    desserts()

  entrees_button = tk.Button(appswind, text="Go to Entrees", font=('Arial', 12), fg='black',
bg='#E16C2E', command=open_entrees)
  entrees_button.place(x=150, y=20 * (len(appslist) + 3))

  desserts_button = tk.Button(appswind, text="Go to Desserts", font=('Arial', 12), fg='black',
bg='#E16C2E', command=open_desserts)
  desserts_button.place(x=300, y=20 * (len(appslist) + 3))

  back_button = tk.Button(appswind, text="Back to Menu", font=('Arial', 12), fg='black',
bg='#E16C2E', command=lambda: [appswind.withdraw(), menu()])
  back_button.place(x=450, y=20 * (len(appslist) + 3))
```

#when researching for the code i discovered that instead of defining a function for each button i can just use the keyword lambda and define the command there itself

```
   Finalise = tk.Button(appswind, text='Finalise order', font=('Arial', 12), fg='black', bg='#E16C2E', command=lambda: [appswind.withdraw(), finaliseorder()])
   Finalise.place(x=300, y=20*(len(appslist)+6))

# ---------- ENTREES WINDOW ----------
def entrees():
   wind3.withdraw()
   entreewind = tkinter.Toplevel(root)
   entreewind.title('ENTREES')
   entreewind.geometry('1500x900')
   entreewind.configure(bg='black')

   t = []

   e = ''
   for i in entreeslist:
      if len(i) > len(e):
         e = i

   # create labels and entry fields for each entree
   for i in entreeslist:
      entry = tk.StringVar()
      entry.set('0')

      j = tk.Label(entreewind, text=i, font=('Arial', 12), fg='white', bg='black')
      j.place(x=0, y=20 * entreeslist.index(i))

      k = tk.Entry(entreewind, textvariable=entry, width=5, font=('Arial', 12))
      k.place(x=7*len(e)+100, y=20 * entreeslist.index(i))

      t.append(entry)
```

```
    # function to save selections
    def entreesfunc():
        global entreesorder, appsorder, dessertsorder
        entreesorder = []
        for i in t:
            entreesorder.append(i.get())

        final = []
        for i in range(len(appslist)):
            if i < len(appsorder) and appsorder[i] != "0":
                final.append(appslist[i] + ", " + appsorder[i])

        for i in range(len(entreeslist)):
            if i < len(entreesorder) and entreesorder[i] != "0":
                final.append(entreeslist[i] + ", " + entreesorder[i])

        for i in range(len(dessertslist)):
            if i < len(dessertsorder) and dessertsorder[i] != "0":
                final.append(dessertslist[i] + ", " + dessertsorder[i])

        global s
        s = ''
        for i in final:
            s = s + '\n' + i
        ordertext = tk.Label(entreewind, text=s, bg='black', fg='white', font=(12))
        ordertext.place(x=800, y=0)

    Submit = tk.Button(entreewind, text='Submit', font=('Arial', 14), fg='black', bg='#E16C2E',
command=entreesfunc)
    Submit.place(x=0, y=20*(len(entreeslist) + 3))

    # buttons to navigate categories
    def open_apps():
        entreewind.withdraw()
        apps()
```

```
    def open_desserts():
        entreewind.withdraw()
        desserts()

    apps_button = tk.Button(entreewind, text="Go to Appetizers", font=('Arial', 12), fg='black',
bg='#E16C2E', command=open_apps)
    apps_button.place(x=150, y=20 * (len(entreeslist) + 3))

    desserts_button = tk.Button(entreewind, text="Go to Desserts", font=('Arial', 12), fg='black',
bg='#E16C2E', command=open_desserts)
    desserts_button.place(x=300, y=20 * (len(entreeslist) + 3))

    back_button = tk.Button(entreewind, text="Back to Menu", font=('Arial', 12), fg='black',
bg='#E16C2E', command=lambda: [entreewind.withdraw(), menu()])
    back_button.place(x=450, y=20 * (len(entreeslist) + 3))

    Finalise = tk.Button(entreewind, text='Finalise order', font=('Arial', 12), fg='black', bg='#E16C2E',
command=lambda: [entreewind.withdraw(), finaliseorder()])
    Finalise.place(x=300, y=20*(len(entreeslist)+6))

# ---------- DESSERTS WINDOW ----------
def desserts():
    wind3.withdraw()
    dessertswind = tkinter.Toplevel(root)
    dessertswind.title('DESSERTS')
    dessertswind.geometry('1500x900')
    dessertswind.configure(bg='black')

    t = []

    e = ''
    for i in dessertslist:
        if len(i) > len(e):
            e = i

    # create labels and entry fields for each dessert
```

```
for i in dessertslist:
    entry = tk.StringVar()
    entry.set('0')

    j = tk.Label(dessertswind, text=i, font=('Arial', 12), fg='white', bg='black')
    j.place(x=0, y=20 * dessertslist.index(i))

    k = tk.Entry(dessertswind, textvariable=entry, width=5, font=('Arial', 12))
    k.place(x=7*len(e)+100, y=20 * dessertslist.index(i))

    t.append(entry)

# function to save selections
def dessertsfunc():
    global dessertsorder, appsorder, entreesorder
    dessertsorder = []
    for i in t:
        dessertsorder.append(i.get())

    final = []
    for i in range(len(appslist)):
        if i < len(appsorder) and appsorder[i] != "0":
            final.append(appslist[i] + ", " + appsorder[i])

    for i in range(len(entreeslist)):
        if i < len(entreesorder) and entreesorder[i] != "0":
            final.append(entreeslist[i] + ", " + entreesorder[i])

    for i in range(len(dessertslist)):
        if i < len(dessertsorder) and dessertsorder[i] != "0":
            final.append(dessertslist[i] + ", " + dessertsorder[i])

    global s
    s = ''
    for i in final:
        s = s + '\n' + i
```

```
        ordertext = tk.Label(dessertswind, text=s, bg='black', fg='white', font=(12))
        ordertext.place(x=800, y=0)

    Submit = tk.Button(dessertswind, text='Submit', font=('Arial', 14), fg='black', bg='#E16C2E',
command=dessertsfunc)
    Submit.place(x=0, y=20*(len(dessertslist) + 3))

    # buttons to navigate categories
    def open_apps():
        dessertswind.withdraw()
        apps()

    def open_entrees():
        dessertswind.withdraw()
        entrees()

    apps_button = tk.Button(dessertswind, text="Go to Appetizers", font=('Arial', 12), fg='black',
bg='#E16C2E', command=open_apps)
    apps_button.place(x=150, y=20*(len(dessertslist)+3))

    entrees_button = tk.Button(dessertswind, text="Go to Entrees", font=('Arial', 12), fg='black',
bg='#E16C2E', command=open_entrees)
    entrees_button.place(x=300, y=20*(len(dessertslist)+3))

    back_button = tk.Button(dessertswind, text="Back to Menu", font=('Arial', 12), fg='black',
bg='#E16C2E', command=lambda: [dessertswind.withdraw(), menu()])
    back_button.place(x=450, y=20*(len(dessertslist)+3))

    Finalise = tk.Button(dessertswind, text='Finalise order', font=('Arial', 12), fg='black',
bg='#E16C2E', command=lambda: [dessertswind.withdraw(), finaliseorder()])
    Finalise.place(x=300, y=20*(len(dessertslist)+6))

# ---------- RUN THE APP ----------
root.mainloop()
mycon.close()
```

ABOUT THE CODE

'''The provided Python code constructs a functional multi-window ordering system for a fictional restaurant, "LEXMI NICOS," utilizing the Tkinter library to manage the graphical user interface. While the application's visual style is admittedly basic due to its reliance on standard Tkinter widgets and absolute positioning (.place), its strength lies in its meticulously organized functional architecture, which ensures a logical, stateful progression from user identification to order finalization. The entire system is held together by a network of global variables that synchronize data across disparate windows, making the complex process of building a multi-category order seamless for the user.

The initial phase of the application focuses on authentication and flow control. The primary root window presents two distinct pathways: Staff or Guest. The staff() function simulates a simple login check against a hardcoded credential set, although its workflow terminates after access is granted or denied. The Guest path, initiated by guest(), immediately prompts the user to declare their membership status, setting the value of the critical global integer variable i (where 1 signifies a member and 0 a non-member). This choice determines the next step: the showmenu() function either sends a non-member directly to the main menu or opens a Member Verification window for members, demanding a passcode check before granting access to the ordering system.

Once the user is authenticated, the Main Menu (menu()) serves as the central navigation hub, presenting the three core categories: Appetizers, Entrees, and Desserts. The interactive ordering process unfolds within three virtually identical functions—apps(), entrees(), and desserts()—each responsible for creating its own dedicated window. Within these screens, menu items are displayed alongside simple tk.Entry widgets where the customer inputs the desired quantity. A crucial architectural decision here is the temporary storage of these inputs: each entry field is bound to a list of local tk.StringVar objects (like tapps) to temporarily hold the numerical input before it’s committed to the permanent record.

The most critical functional element of the code is the submission mechanism (e.g., the appsfunc within apps()). When the user clicks Submit, this function first retrieves all quantity values from the

temporary local strings and uses them to overwrite the permanent global order list for that specific category (e.g., appsorder). Immediately after, the function performs a vital synchronization step: it iterates through all three global order lists (appsorder, entreesorder, and dessertsorder) and rebuilds the master global final list. This comprehensive list aggregates every item and quantity selected across the entire menu, thereby providing a constantly updated live order preview on the right side of the active category window. This synchronization guarantees that the order state is never lost, regardless of which category screen the user is viewing.

Finally, the finaliseorder() function brings the process to a close. This screen serves as the definitive Order Summary, recalculating the final list one last time to ensure absolute accuracy. Every selected item and quantity is displayed for review. Importantly, this final window retains the same navigation buttons as the category screens. This design feature provides a critical error correction loop: should the customer spot a mistake, they can simply click "Go to Entrees," adjust the quantity, click Submit to update the global lists, and return to Finalise Order, where the changes are reflected instantly. This structure ensures that the entire order compilation is flexible, persistent, and responsive before the customer physically confirms their selection.'''

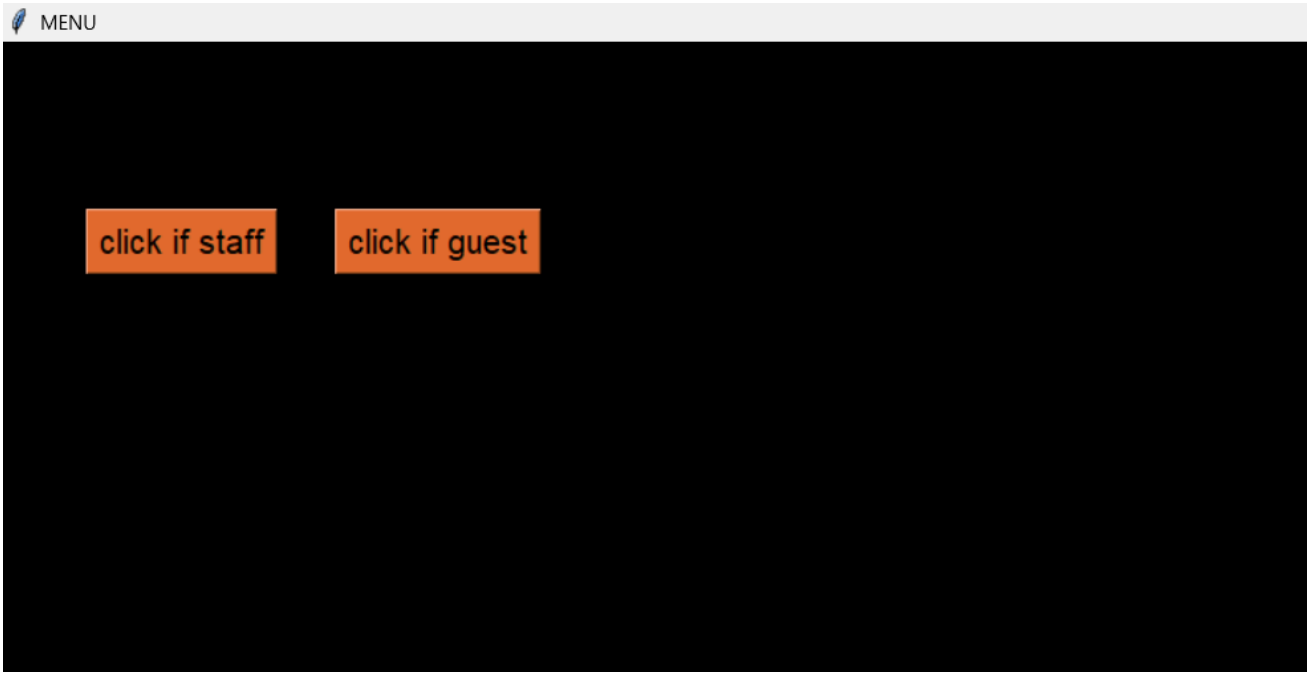
DATABASE STRUCTURE

```
mysql> use lexminicos;
Database changed
mysql> select*from staff;
+---------+----------+
| name    | password |
+---------+----------+
| DEVESHH | 1        |
| SURESH  | 2        |
+---------+----------+
2 rows in set (0.00 sec)

mysql> select*from members;
+---------+----------+
| name    | password |
+---------+----------+
| DEVESHH | 1        |
| SURESH  | 2        |
+---------+----------+
2 rows in set (0.00 sec)

mysql> desc staff;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| name     | varchar(20) | YES  |     | NULL    |       |
| password | varchar(10) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
2 rows in set (0.06 sec)
```

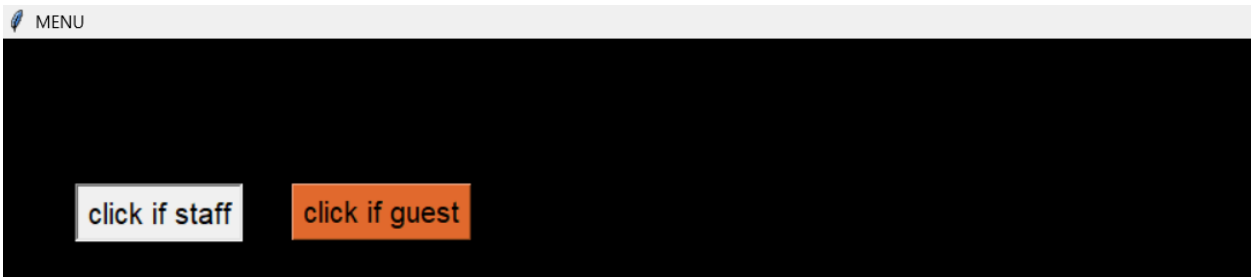
```
mysql> desc members;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| name     | varchar(20) | YES  |     | NULL    |       |
| password | varchar(10) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

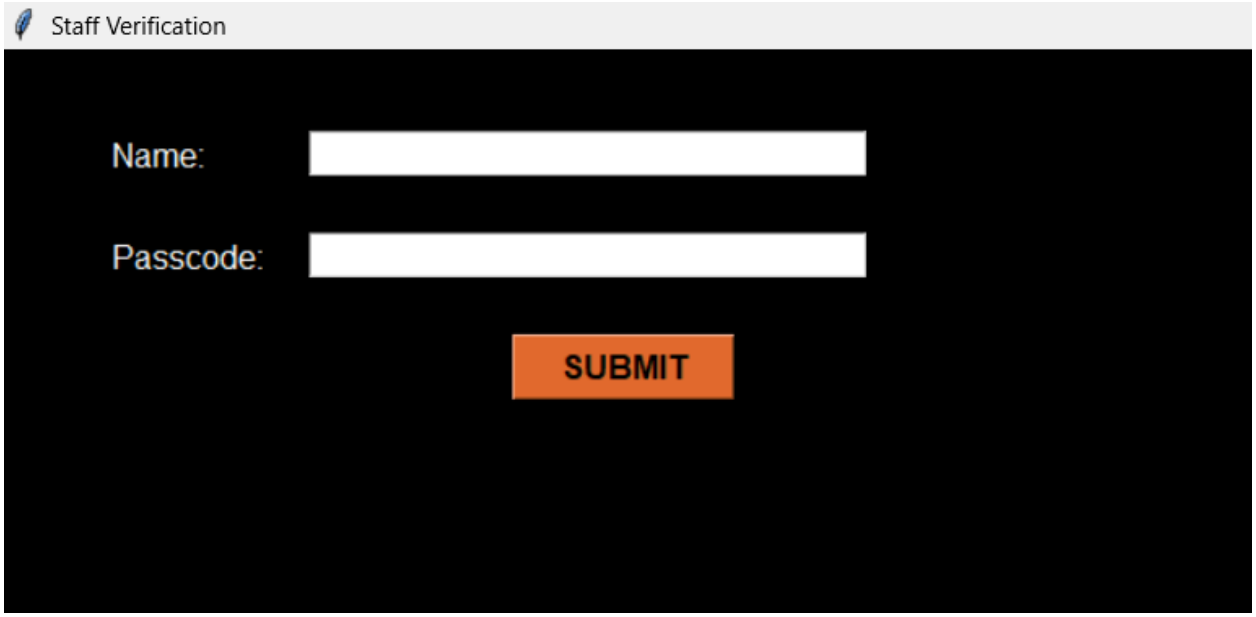
OUTPUT

As soon as program starts running

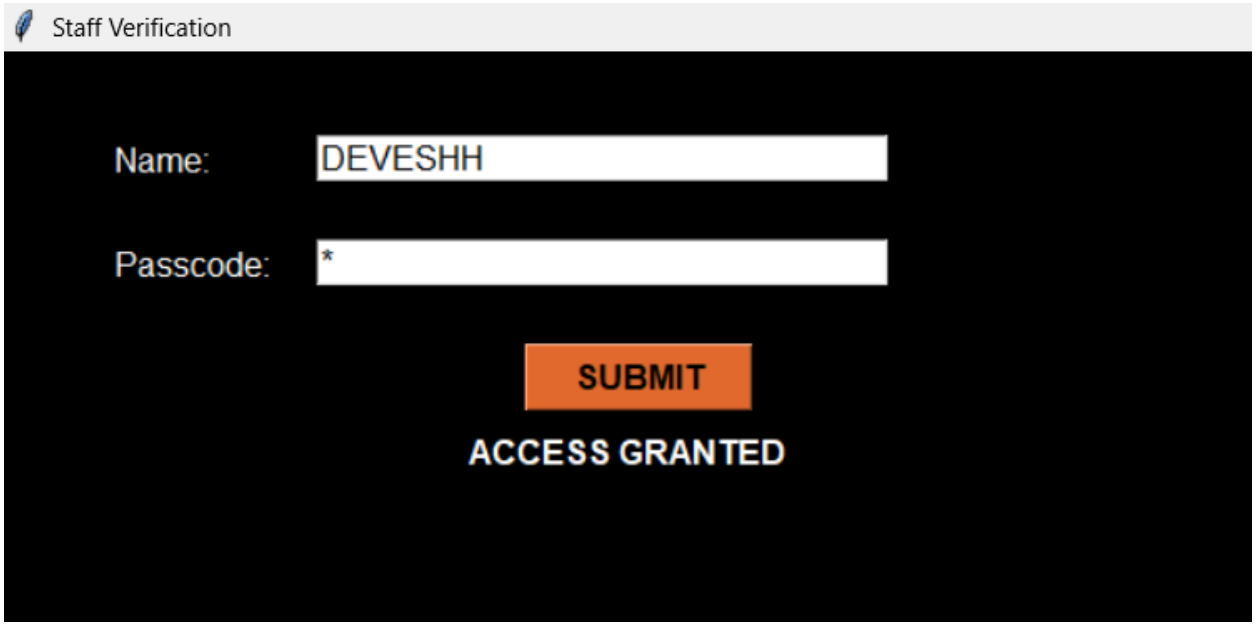


1. Staff uses it(click if staff is clicked)

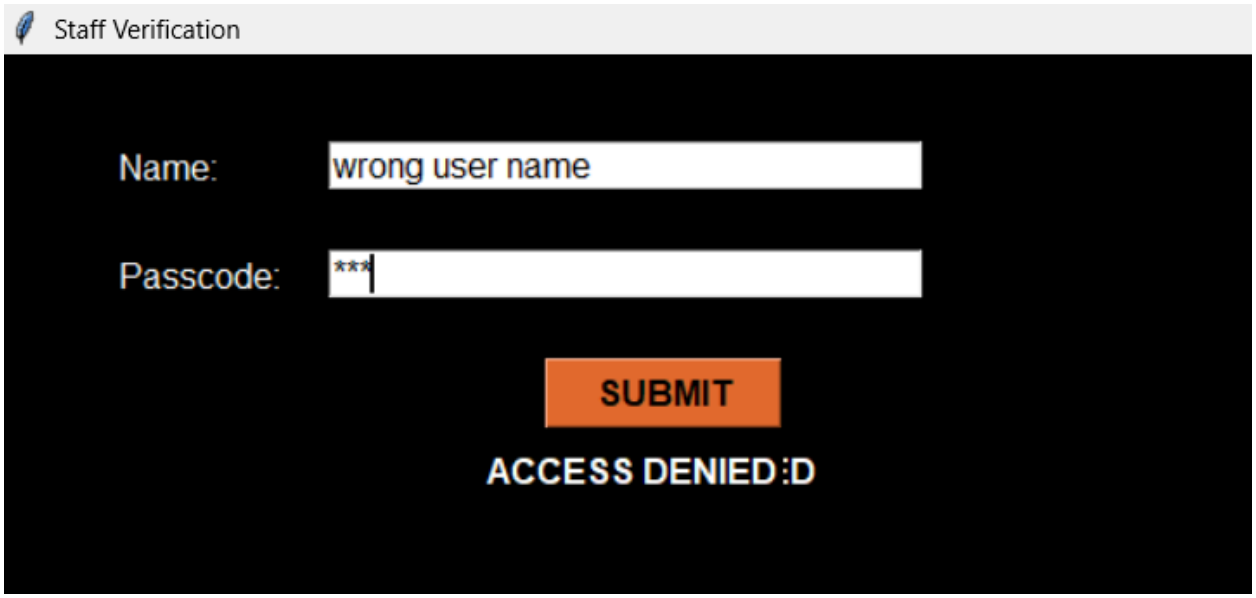




If correct user name and password is selected the below image shows what happens

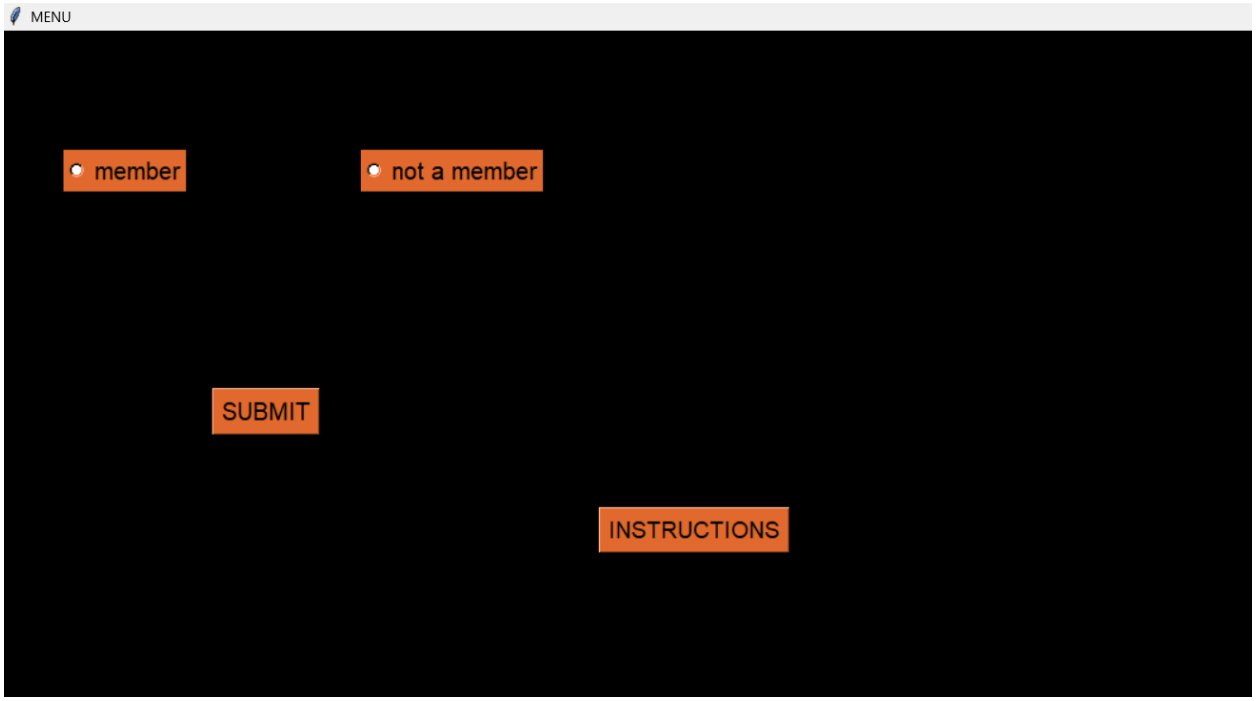


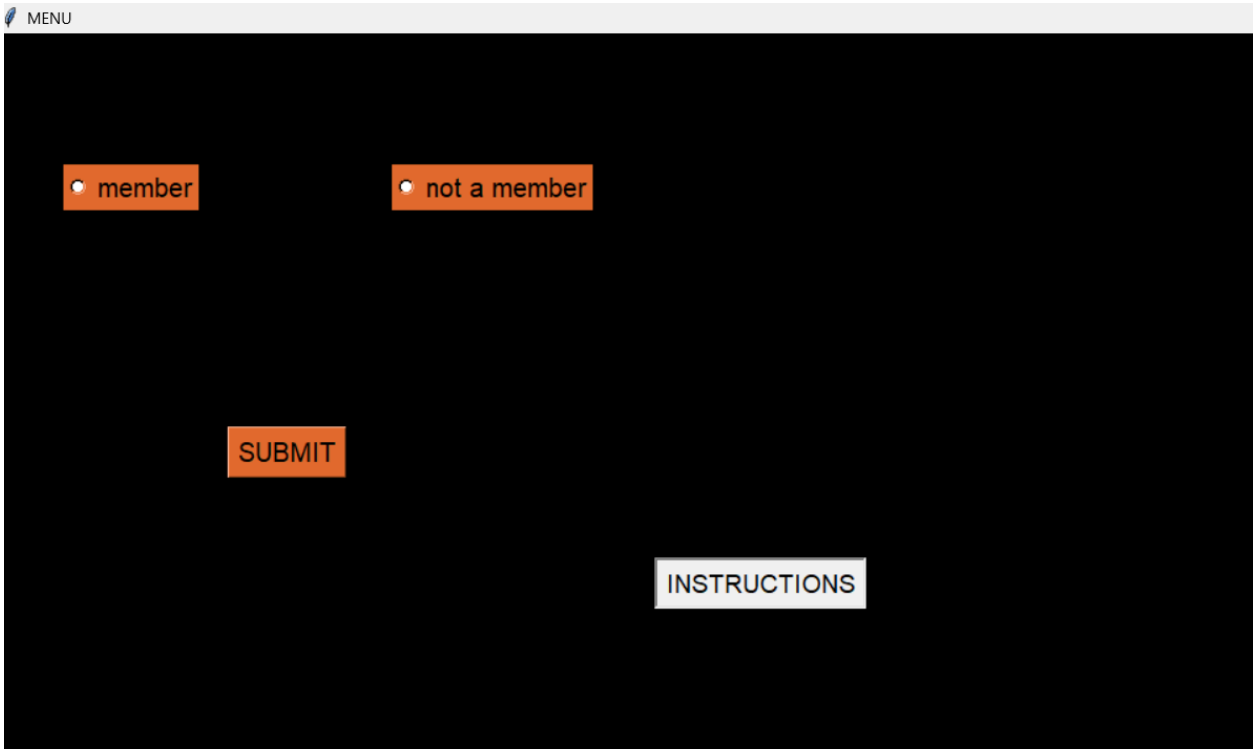
If incorrect username password combination is entered the following image illustrates what happens

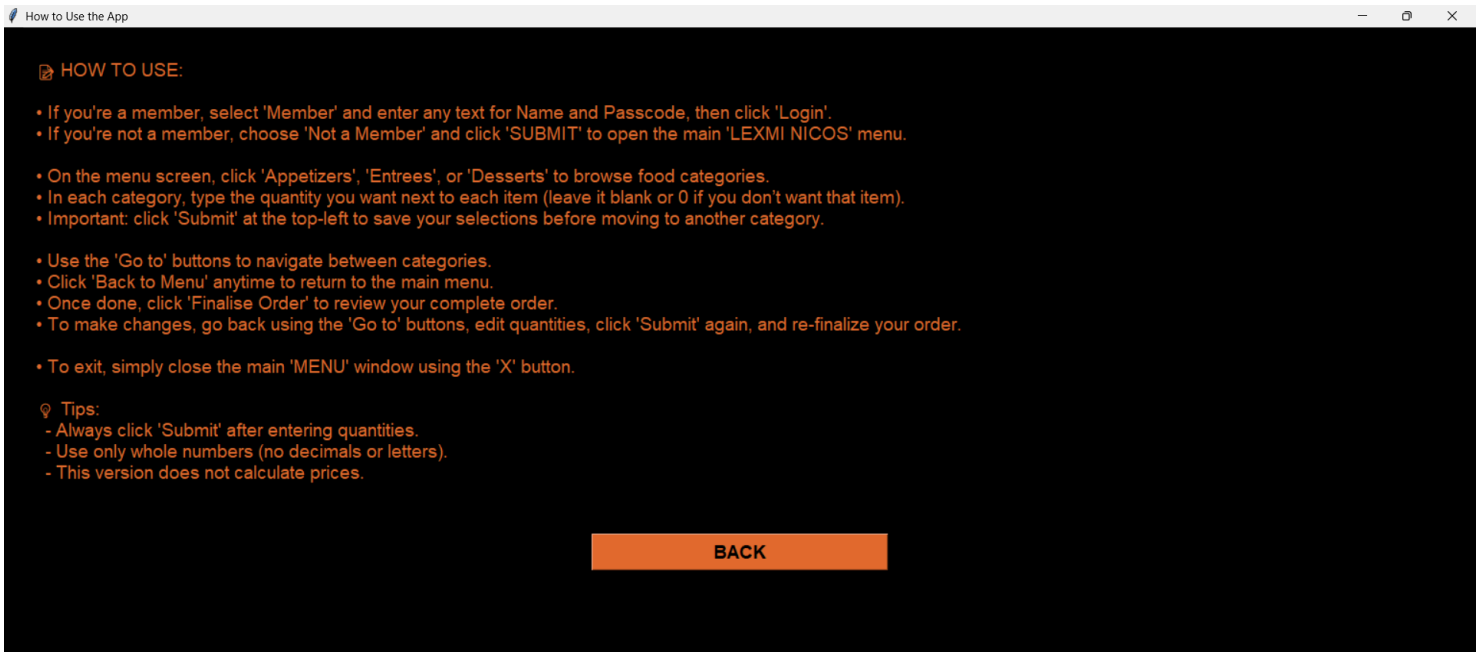

Staff Verification
Name:
wrong user name
Passcode:
***
SUBMIT
ACCESS DENIED :D

2. If guest

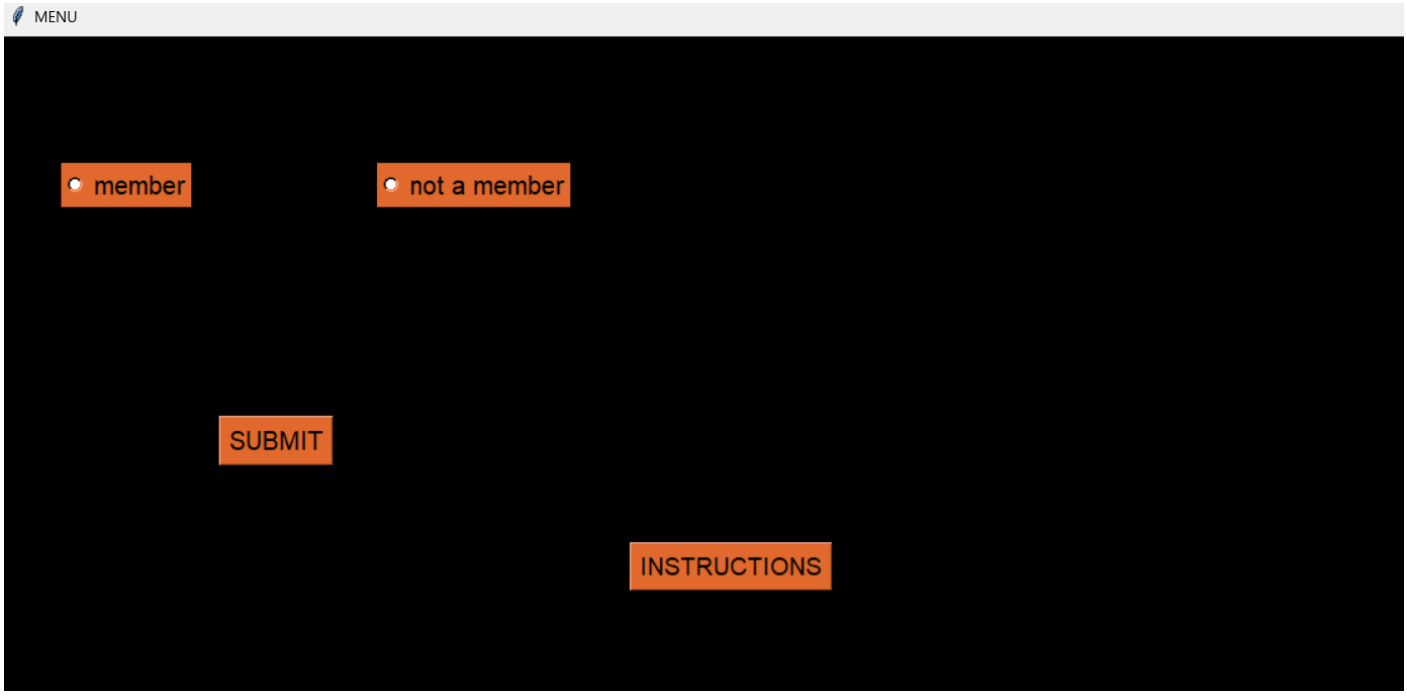

MENU
click if staff
click if guest


MENU
member
not a member
SUBMIT
INSTRUCTIONS


MENU
member
not a member
SUBMIT
INSTRUCTIONS

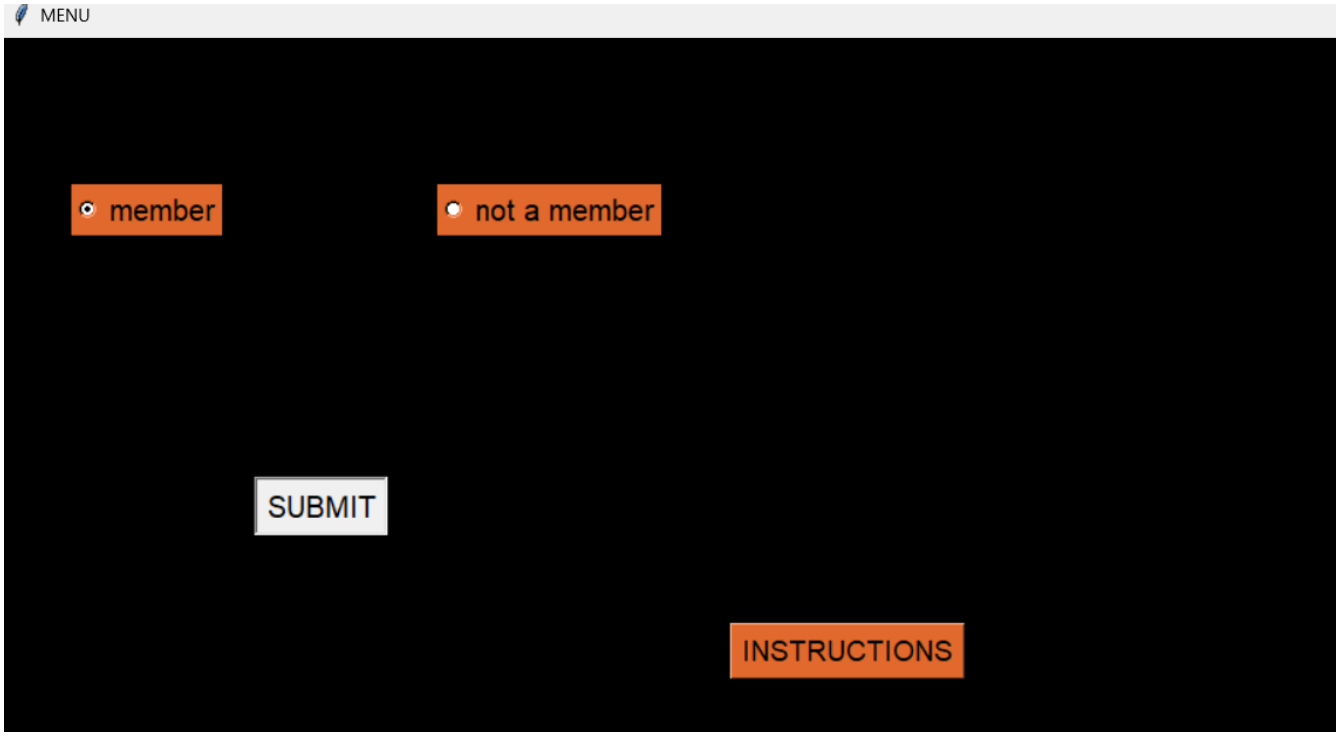


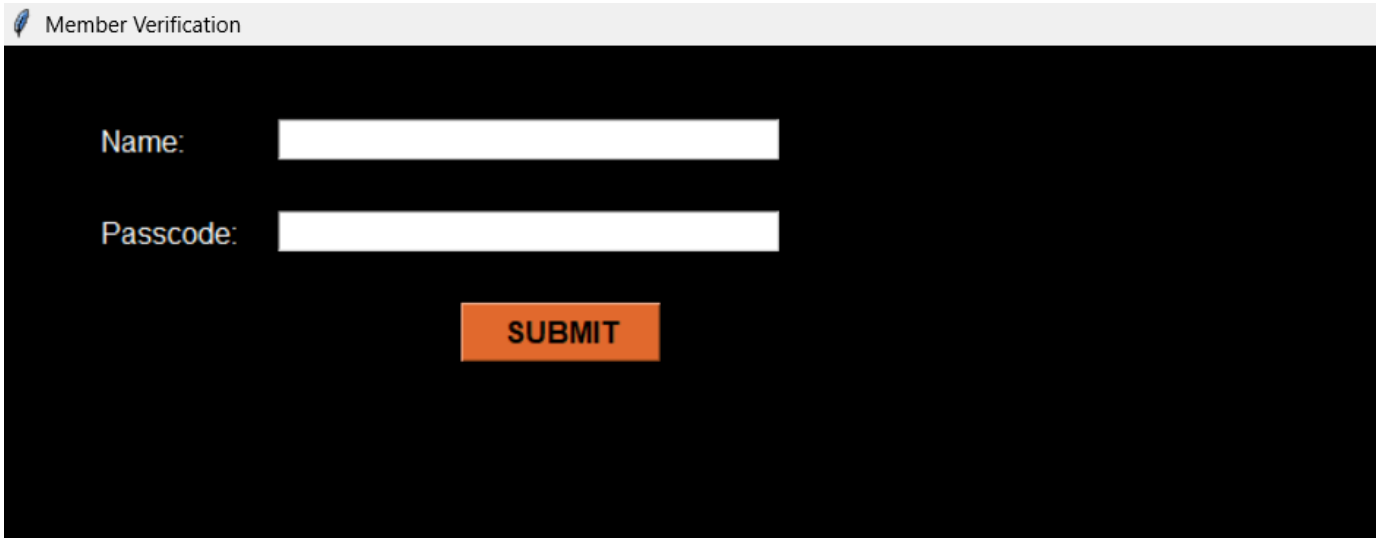
Clicking instructions opens the instructions menu



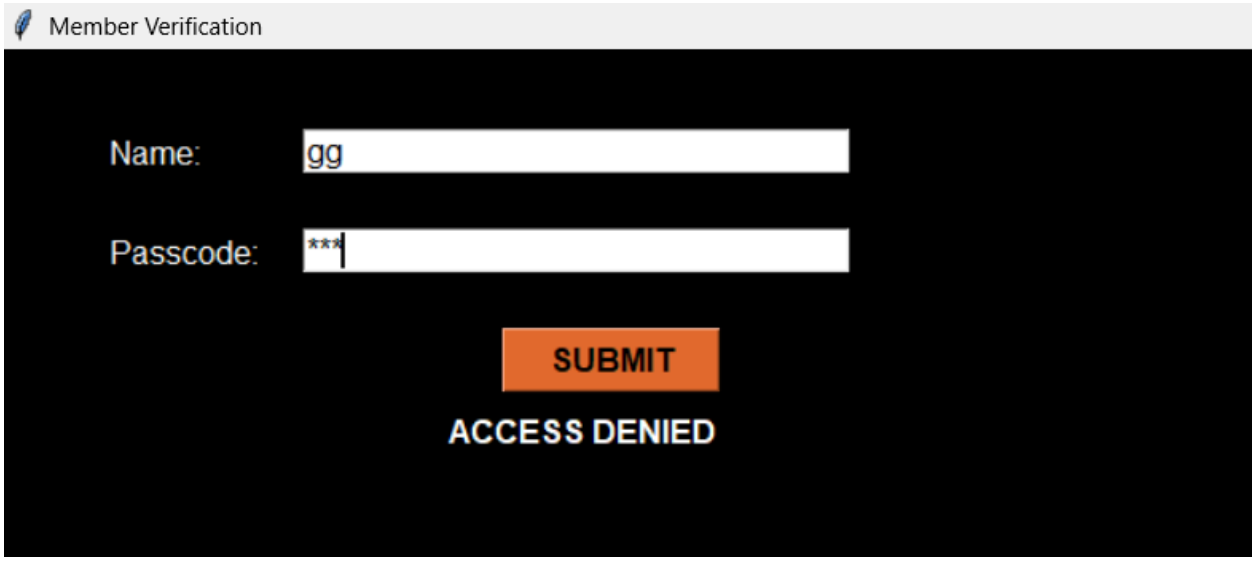
Clicking back returns to the previous window

If member:
Click the correct radiobutton and give submit

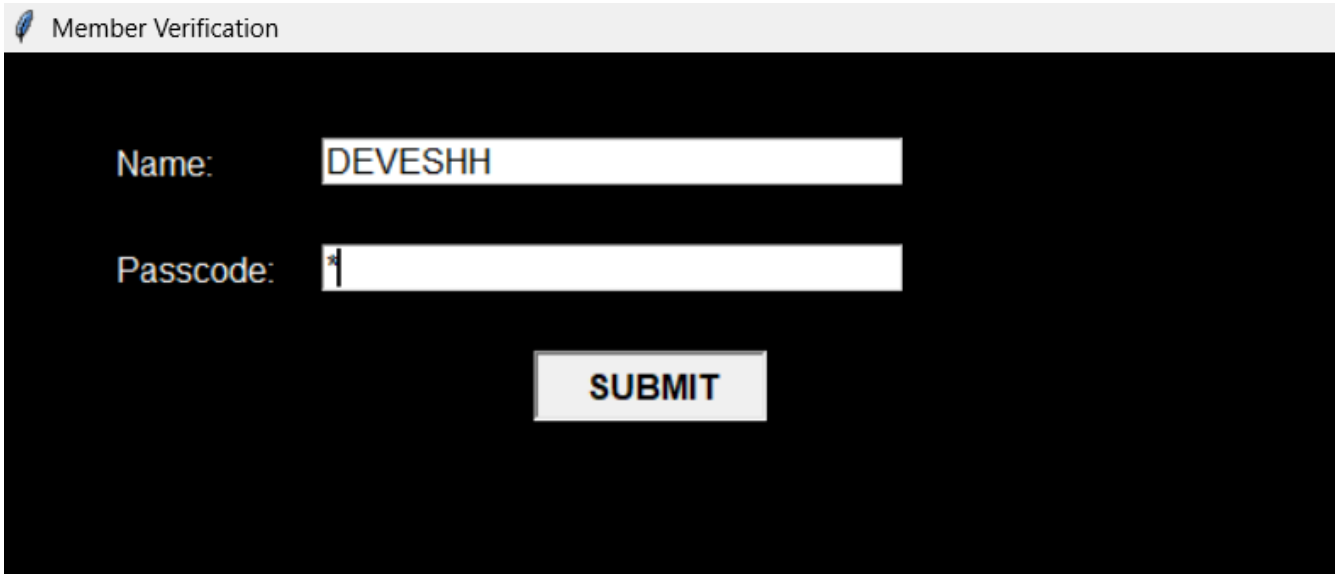




If incorrect details selected

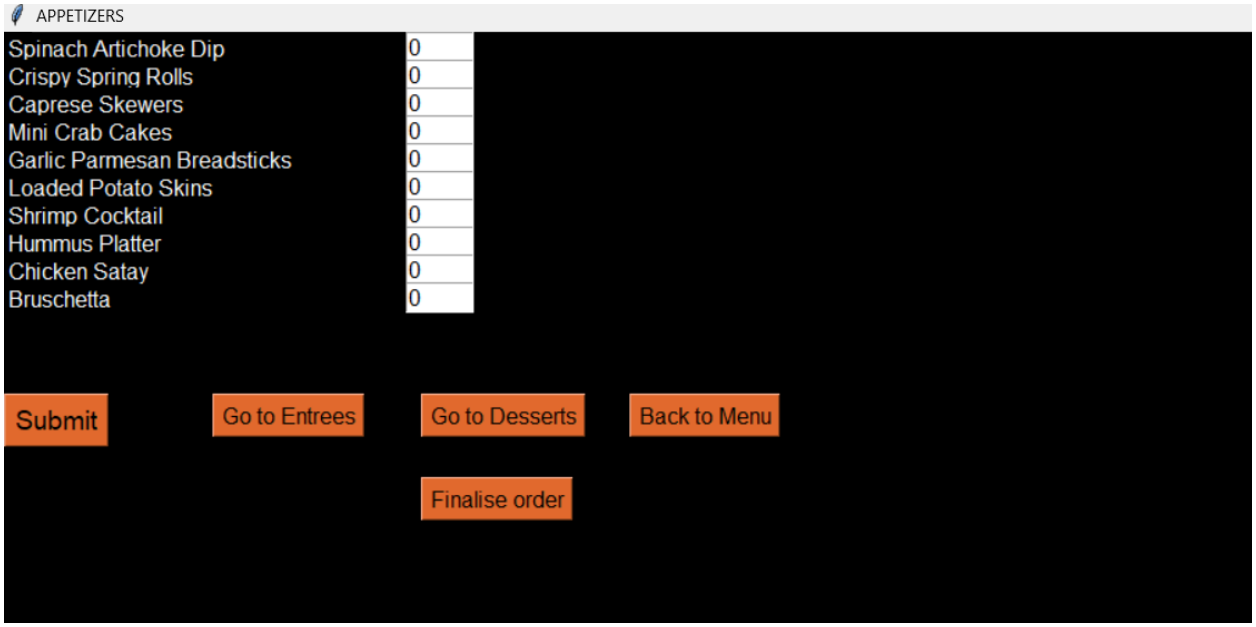


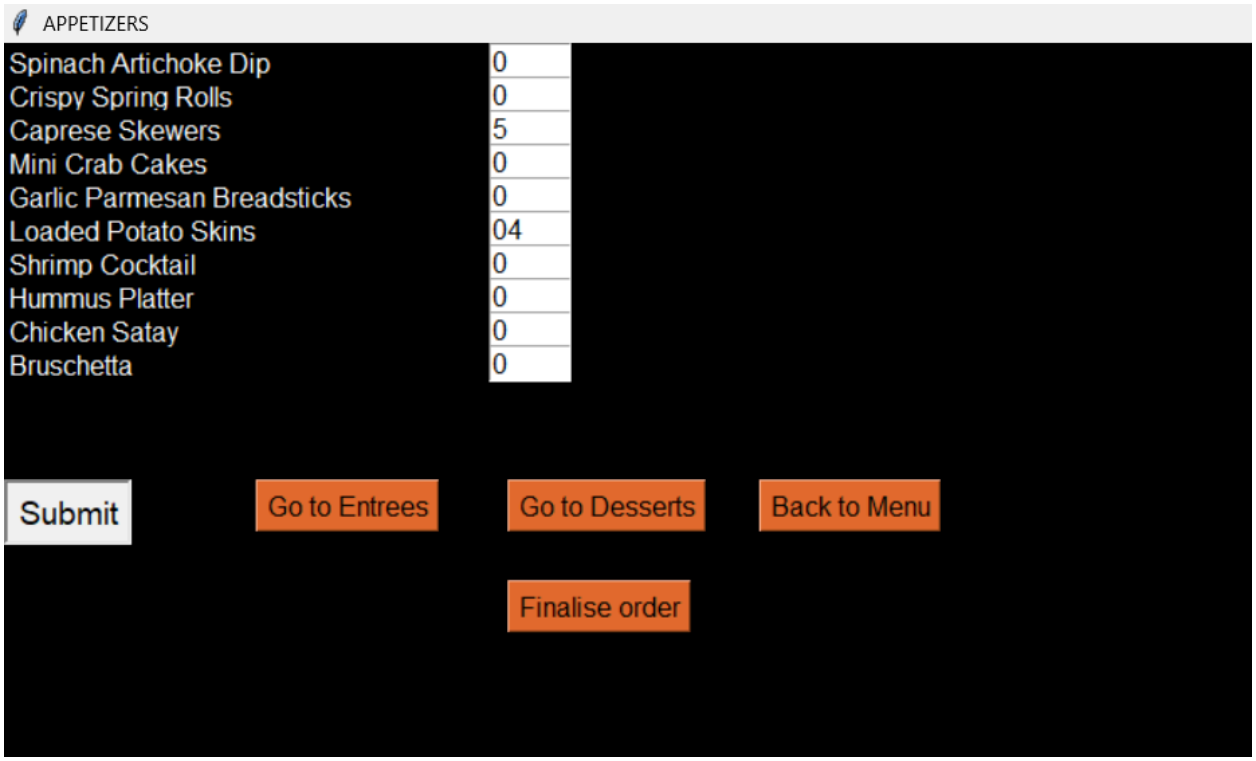
If correct details have been entered the following 2 images show what happens:



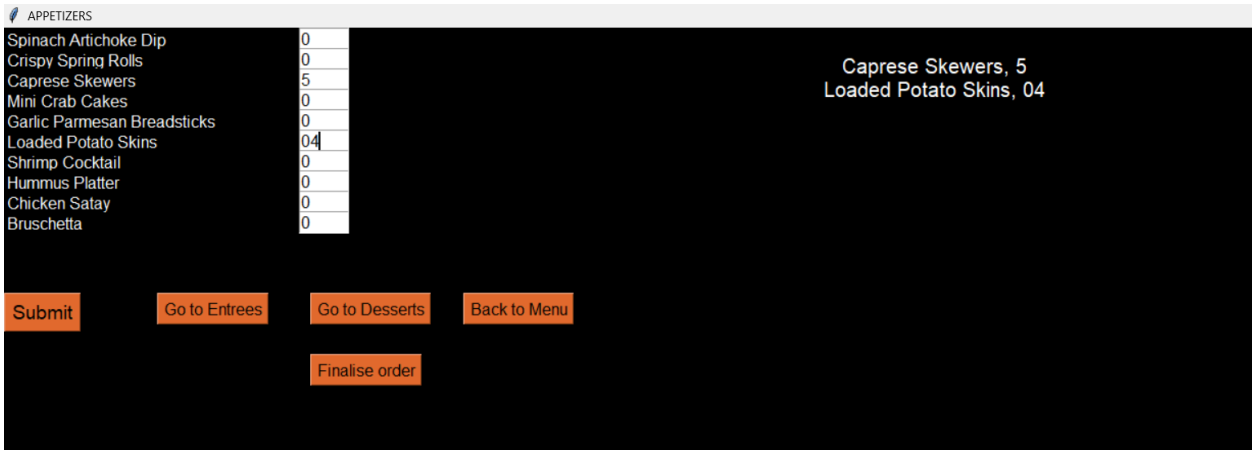
If directly not a member is selected the following image is shown:

Appetizers is chosen

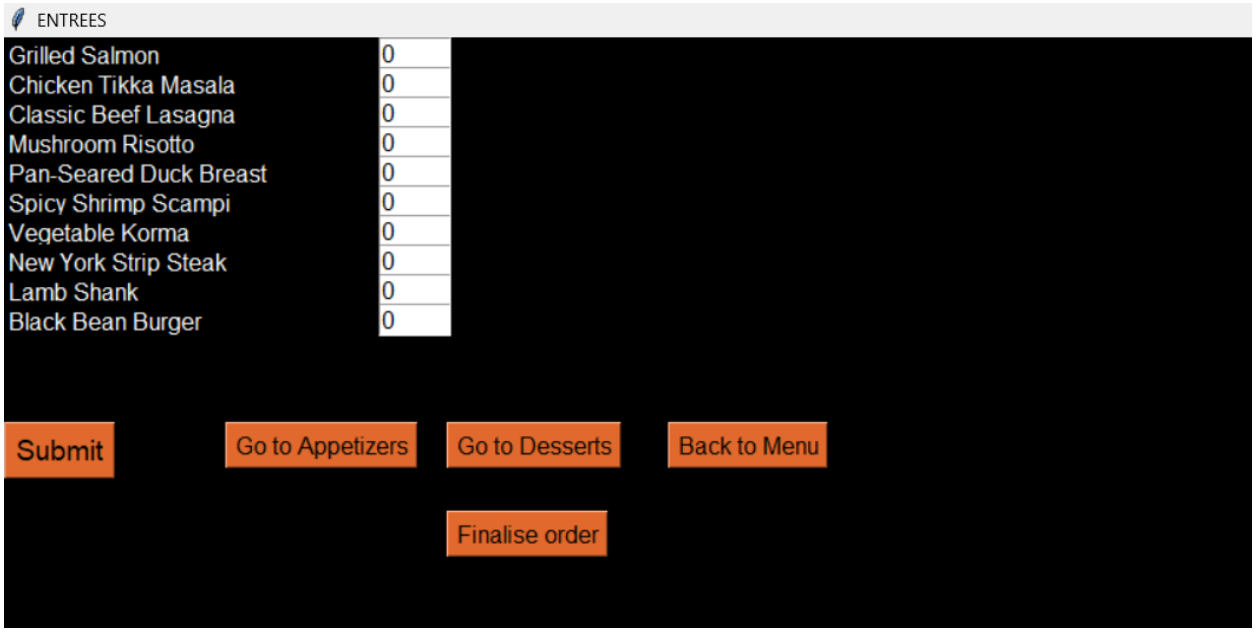




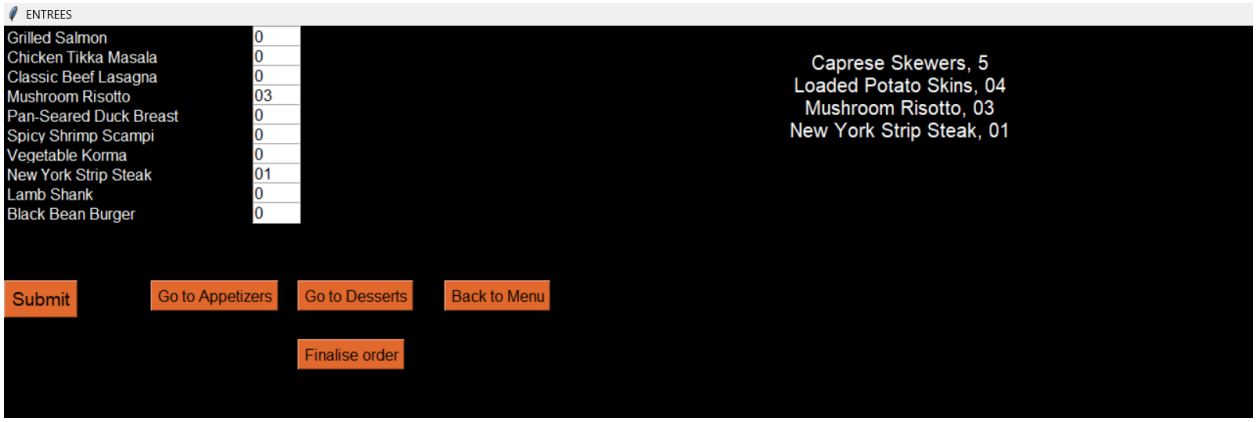
Submit is clicked then the following image illustrates what happens



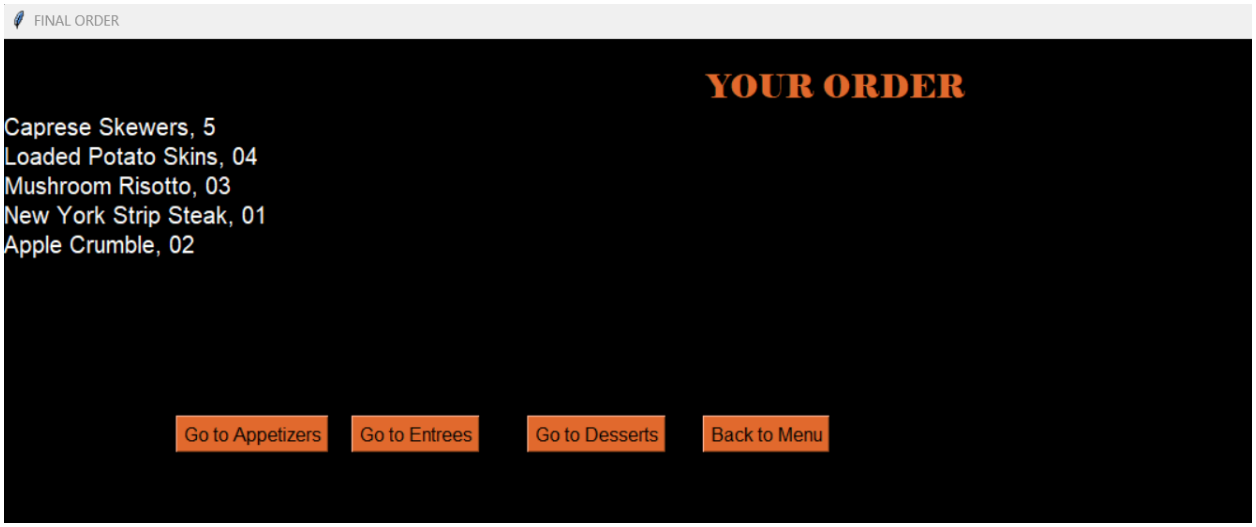
If entrees is clicked then the following image illustrates what happens



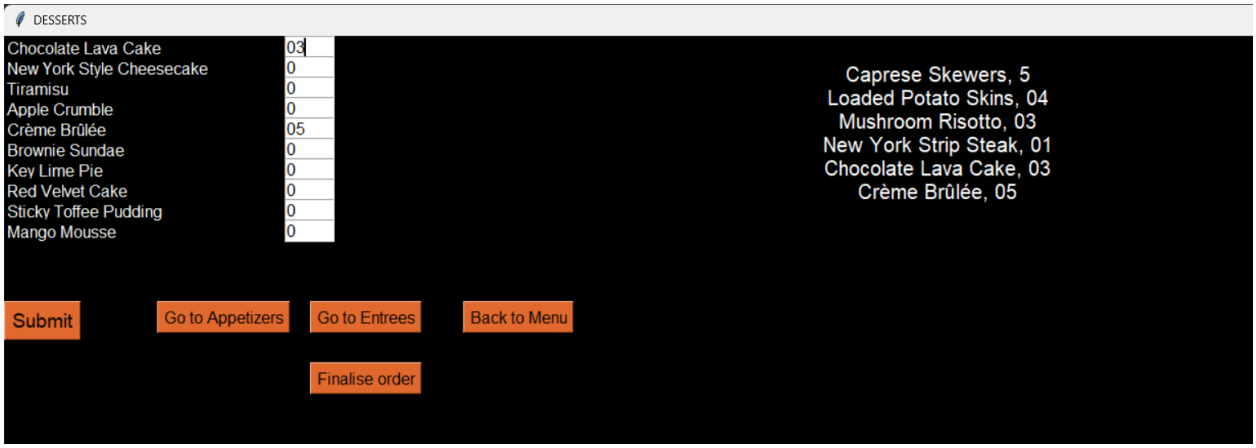
Wanted quantities are selected and submitted

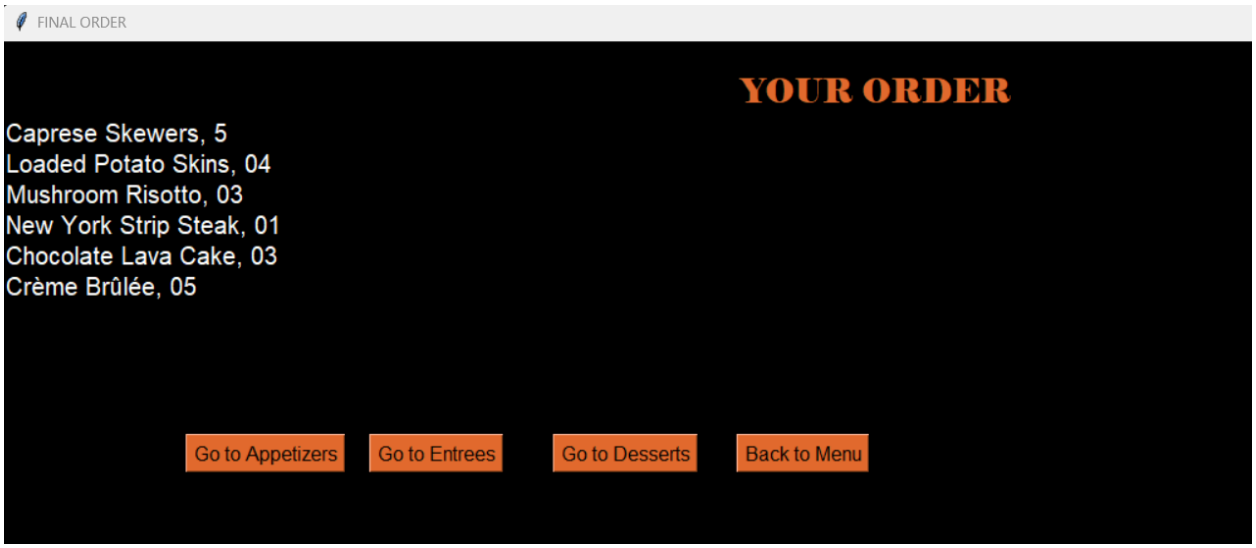


after all is selected finalize order is given



If in case there is any change to be made then u can click that section and change your order for example hear the desserts have been changed





SCOPE FOR ENCHANCEMENTS:

1. Clicking of the staff menu can be linked up to a hotel management system
2. Order from users can be assigned a table number but for this project since it is used locally this has not been done
3. Automation of submit button and a feature to call the waiter incase of any problems

CONCLUSION

The *Lexmi Nicos Order Suite* stands as a refined integration of technology and service, combining the reliability of MySQL with the elegance of a graphical interface. Designed with clarity, precision, and poise, the system ensures seamless order placement while reflecting the sophistication of a well-orchestrated service experience. Beyond its immediate functionality, the project demonstrates how thoughtful design and disciplined engineering can elevate even routine tasks into an interface of distinction. In essence, this suite is not merely a program but a statement of how technology, when crafted with elegance, can serve with the grace of fine service.

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4. https://youtu.be/nwht_3zUGI0?si=V_ZY3zKk3fo1feY0
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